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Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS



Half Past Paper – Week 16

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
2a	Venn diagrams with set notation	3	U748
2b	Venn diagrams with set notation	2	U748
4a	Graphs of cubic functions	2	U980
4b	Graphs of cubic functions	2	U980
6	Multiplying column vectors by a scalar	2	U564
8a	Finding original values in percentage calculations	2	U286
8b	Compound interest calculations	3	U332
10a	Index rules with positive indices, Simplifying expressions using index laws	1	U235,U662
10b	Simplifying algebraic fractions by factorising into one bracket	1	U437
10c	Simplifying expressions using index laws	2	U662
12a	Finding equations of linear real-life graphs	2	U862
12b	Finding equations of linear real-life graphs	1	U862
14	Constructing and solving equations	4	U599
16	Position-to-term rules for quadratic sequences	3	U206
18	Finding the surface area of cones and frustums, Similar shapes	4	U523, U334, U578,U110
20a	Probabilities as fractions, Constructing and solving equations, Writing algebraic proofs	3	U408, U599, U582
20b	Tree diagrams for dependent events	5	U729
22	Equations of circles and tangents	5	U567
	Total	47	

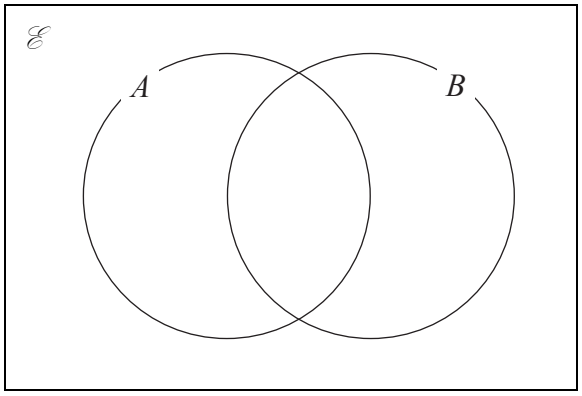
Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{\text{even numbers}\}$
$B = \{\text{factors of } 10\}$

(*a*) Complete the Venn diagram for this information.



(3)

A number is chosen at random from the universal set, $\mathscr{E}$

(*b*) Find the probability that this number is in the set $A \cap B$

.....

(2)

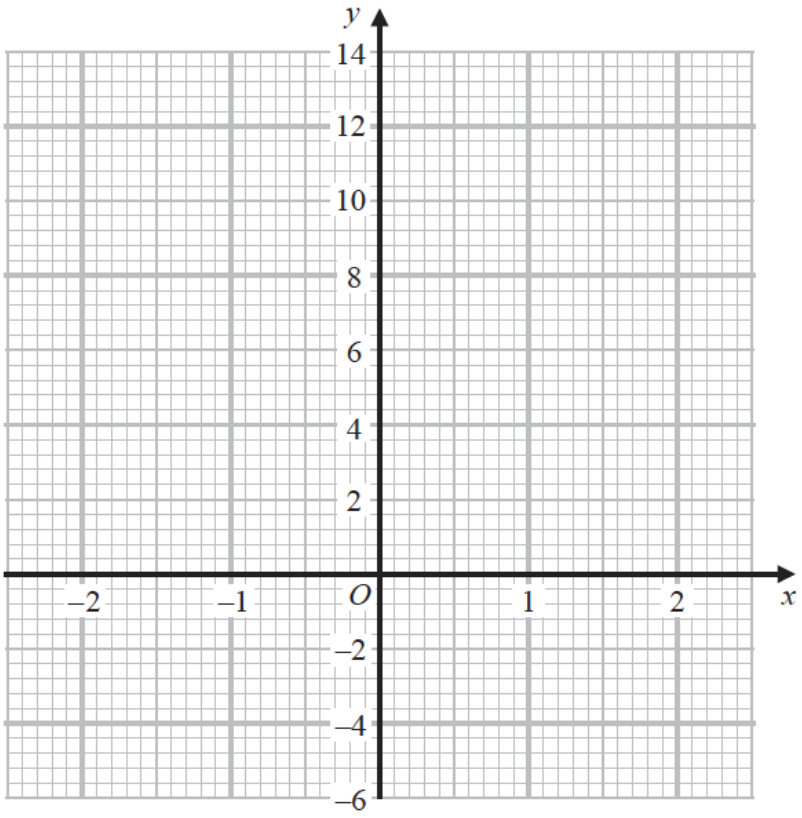
(Total for Question 2 is 5 marks)

4 (*a*) Complete the table of values for $y = 5 - x^3$

x	–2	–1	0	1	2
y		6			

(2)

(*b*) On the grid below, draw the graph of $y = 5 - x^3$ for values of x from –2 to 2



(2)

(Total for Question 4 is 4 marks)

6 $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$

Find $2\mathbf{a} - 3\mathbf{b}$ as a column vector.

$\begin{pmatrix} \ldots\ldots \\ \ldots\ldots \end{pmatrix}$

(Total for Question 6 is 2 marks)

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8 Tariq buys a laptop.
He gets a discount of 5% off the normal price.
Tariq pays £551 for the laptop.

(*a*) Work out the normal price of the laptop.

£.....
(2)

Joan invests £6000 in a savings account.
The savings account pays compound interest at a rate of

2.4% for the first year
1.7% for each extra year.

(*b*) Work out the value of Joan's investment at the end of 3 years.

£.....
(3)

(Total for Question 8 is 5 marks)

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10 (*a*) Simplify $\left(\frac{1}{m^2}\right)^0$

.....

(1)

(*b*) Simplify $\frac{8(x-4)}{(x-4)^2}$

.....

(1)

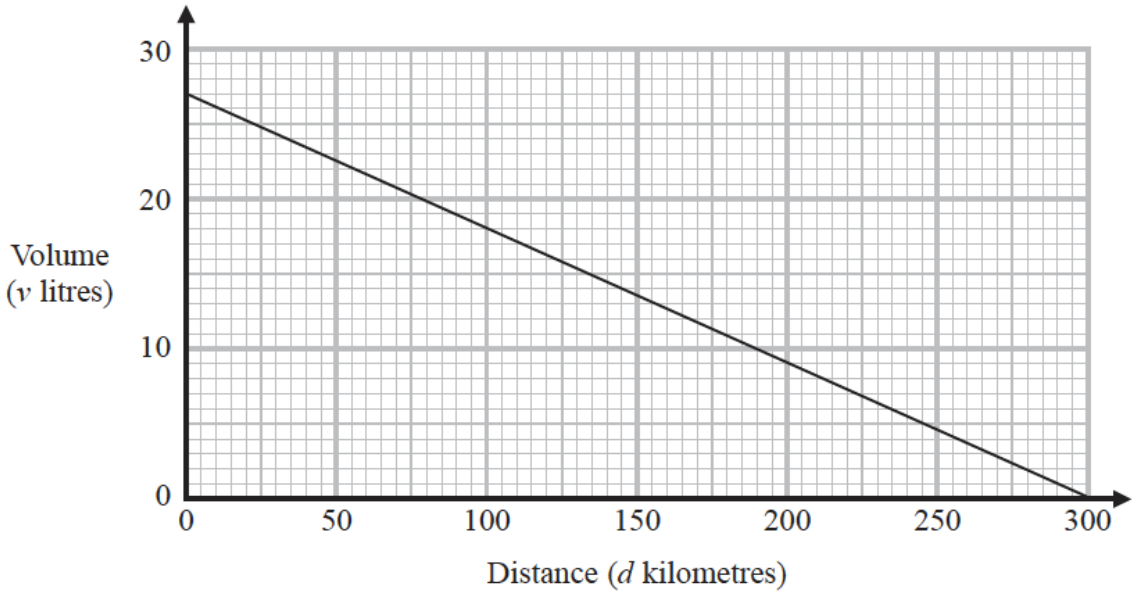
(*c*) Simplify $(3n^4w^2)^3$

.....

(2)

(Total for Question 10 is 4 marks)

12 The graph gives information about the volume, v litres, of petrol in the tank of Jim's car after it has travelled a distance of d kilometres.



(a) Find the gradient of the graph.

.....

(2)

(b) Interpret what the gradient of the graph represents.

.....

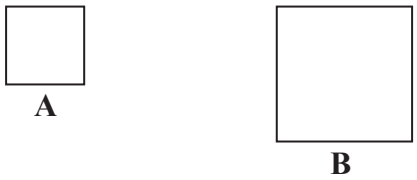
.....

.....

(1)

(Total for Question 12 is 3 marks)

14 Here are two squares, **A** and **B**.



The length of each side of square **B** is 4 cm greater than the length of each side of square **A**.
The area of square **B** is 70 cm^2 greater than the area of square **A**.

Find the area of square **B**.
Give your answer correct to 3 significant figures.
You must show all your working.

..... cm^2

(Total for Question 14 is 4 marks)

16 Here are the first five terms of a quadratic sequence.

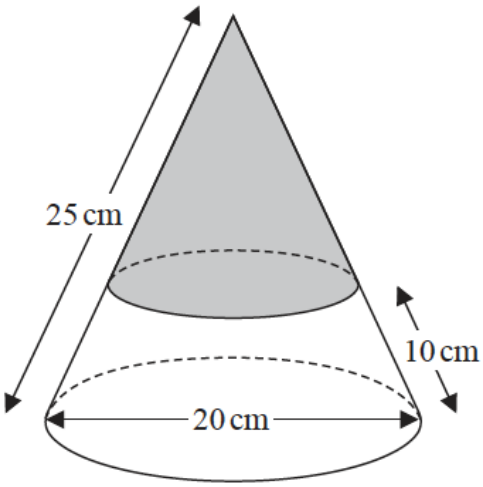
10 21 38 61 90

Find an expression, in terms of n, for the nth term of this sequence.

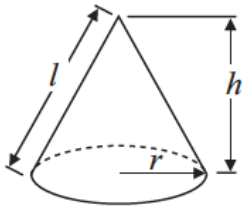
.....

(Total for Question 16 is 3 marks)

18 The diagram represents a solid cone.



Curved surface area of cone = πrl



The cone has a base diameter of 20 cm and a slant height of 25 cm.

A circle is drawn around the surface of the cone at a slant height of 10 cm above the base.
The curved surface of the cone above the circle is painted grey.

Work out the area of the curved surface of the cone that is **not** painted grey.
Give your answer as a multiple of π
You must show all your working.

..... cm^2

(Total for Question 18 is 4 marks)

20 There are only red sweets and yellow sweets in a bag.

There are n red sweets in the bag.
There are 8 yellow sweets in the bag.

Sajid is going to take at random a sweet from the bag and eat it.

He says that the probability that the sweet will be red is $\frac{7}{10}$

(*a*) Show why the probability cannot be $\frac{7}{10}$

(3)

After Sajid has taken the first sweet from the bag and eaten it, he is going to take at random a second sweet from the bag.

Given that the probability that both the sweets he takes will be red is $\frac{3}{5}$

(*b*) work out the number of red sweets in the bag.
You must show all your working.

.....

(5)

(Total for Question 20 is 8 marks)

22 **C** is a circle with centre the origin.

A tangent to **C** passes through the points (–20, 0) and (0, 10)

Work out an equation of **C**.
You must show all your working.

.....

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER IS 47 MARKS